MINOR SPECIALISATION PROJECT · REVIEW OF MEASURED RESULTS

Cross-day verification went from one user refused 8 of 10 times to zero rejections

An evidence-first review of the **gait-biometrics** pipeline. Overnight, a multisensor descriptor plus lagged-covariance backend cut the frozen cross-day error from 7.0% EER with a 22% genuine-reject rate to 0.0% EER with no rejections — but that headline is **exploratory reuse of five already-inspected users** and the false-accept rate still sits at 1.5%, above target. A separate pooled-calibration study also improved the development false-accept rate. Every number is reproduced from committed report artefacts, negative results are kept visible, and reused test cohorts are labelled as such.

- **0** genuine rejections at the new frozen threshold
- **3** datasets · UCI HAR · WISDM AR v1.1 · Santos v7
- **exploratory** cross-day test reuses 5 inspected users
- **1** negative result retained in full
- source commit **4014d85**
- 24 October 2026

WISDM VERIFICATION EER

9.58%

▼ 52.3% vs 20.10% baseline

locked test · ROC AUC 0.9622 · 10,000 comparisons

CROSS-DAY EER (SANTOS V7)

0.0%

▲ from 7.0% — the 8/10 user is fixed

exploratory · 5 reused test users · AUC 1.0 · FAR 1.5%

POOLED CALIBRATION FAR

0.836%

▼ from 0.917% · FRR 19.3→15.1%

policy C · 12-person refit · DEV-only

PREPROCESSING SPEED-UP

15.72×

bit-identical output

100,000 rows → 1,561 windows

MULTISENSOR DEV EER

6.68%

▼ from 11.74% accel-only control

accel+gyro+invariant fusion · 5 folds

HAR ACTIVITY ACCURACY

90.53%

93.69% under the older selection rule

multitask track · 2,947 test windows

01 · VERIFICATION

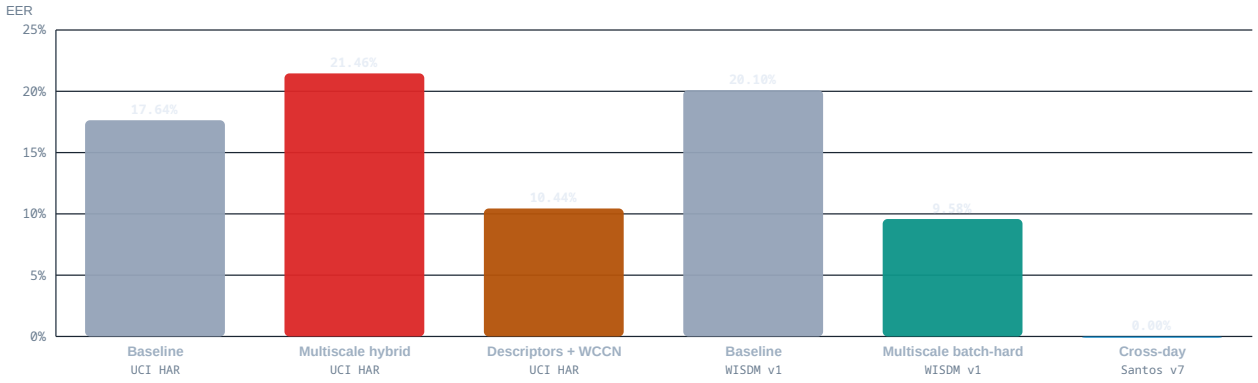
How far the identity check has come — and where it still fails

Within-recording verification on held-out subjects: 10,000 balanced same-activity comparisons per configuration (seed 42), thresholds selected on the validation split and then frozen, and EER reported as an independent quantity rather than as an alias of the operating point. Cross-day results live in section 04 because they use a different cohort and dataset.

Verification error by configuration

Lower is better for EER, FAR and FRR; higher is better for AUC. Cross-day row is a different dataset — not a control for the others.

EER | ROC AUC | Fixed FAR | Fixed FRR



frozen baseline · selected new method · negative result (validation winner that failed to generalise) · exploratory follow-up (test cohort already seen) · cross-day study (separate dataset)

Locked test results, side by side

HAR and WISDM share the within-recording protocol. Evidence status is part of the result, not a footnote.

DATASET	METHOD
UCI HAR	Frozen baseline (20 ep)
UCI HAR	Multiscale hybrid ensemble
UCI HAR	Descriptors + WCCN (0.2)
WISDM AR v1.1	Frozen baseline (20 ep)
WISDM AR v1.1	Multiscale batch-hard ense

Read the fixed FRR column as a usability number, not a ranking. The HAR descriptor follow-up has the best error rate (10.44%) and the lowest false-accept rate (0.54%), yet still refuses 3 in 10 genuine attempts. The comparison is not architecture-controlled: the neural bundle changes normalisation, sampling, objective, augmentation, 40 vs 20 epochs and 3-member vs single-model inference at once.

Five-window prefix enrollment

A separate protocol: five enrollment windows per identity, two skipped, every surviving probe compared against the template. Not directly comparable with random-pair EER.

WISDM METHOD	EER	A
Frozen baseline	28.37%	0.78
Selected neural ensemble	20.42%	0.91

Enrollment improved, but it is not a session protocol. Index purging assumes preserved within-subject order; raw recording offsets are unavailable, so raw non-overlap and independent enrollment/probe sessions are **not** established. HAR enrollment *worsened* for the primary neural winner and is reported in full in the repository rather than dropped.

What is established so far: on WISDM walking, a validation-locked ensemble halves the error rate of the previous baseline and keeps the false-accept rate at 8.26% at a frozen threshold. **What is not:** a deployable operating point, phone-based authentication, or continuous passive authentication.

02 · MODEL DEVELOPMENT

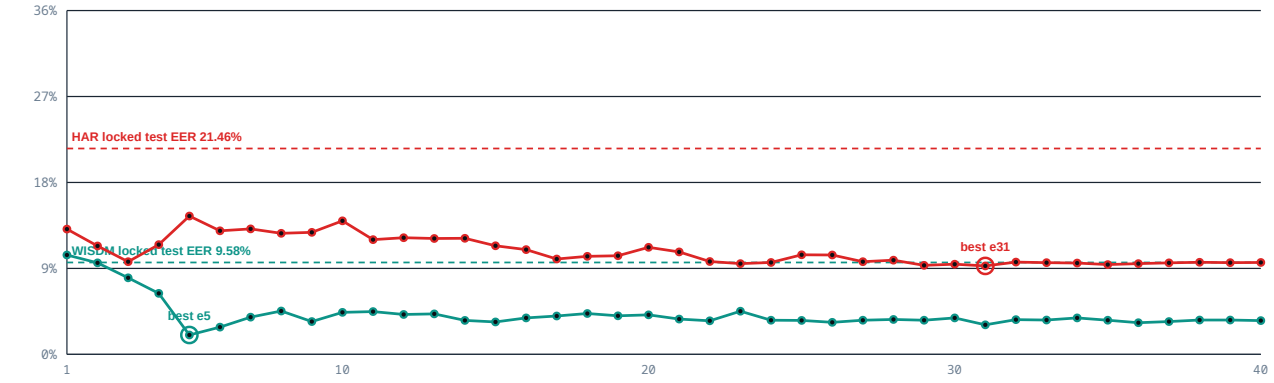
Selection discipline: validation chooses, test only reports

Six neural methods were trained for 40 epochs at three seeds each, with three descriptor baselines, giving 15 validation candidates per dataset. The winner was chosen on validation EER, its artefacts were hashed and locked, and the test split was read once. The curves below show how much of the error was visible before the test split was touched — and how much of it appeared afterwards.

Validation trajectory of the selected methods (40 epochs)

Dashed lines are the locked test EER of the same method. The gap between the two is the generalisation gap the study exists to expose.

Val EER | Val AUC | Loss

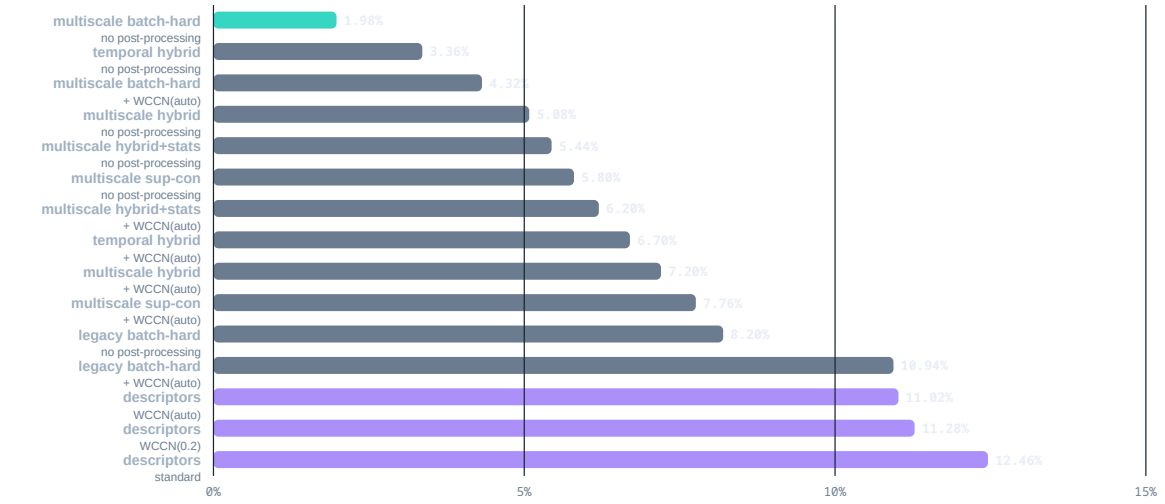


WISDM v1 · multiscale batch-hard (selected) UCI HAR · multiscale hybrid (selected, then failed)

All 15 validation candidates

Selection happened here, on validation, before any test access. The winner is highlighted.

WISDM v1 | UCI HAR



selected winner | other candidate | descriptor baseline

What the protocol guarantees

- **Selection on validation only.** No seed, threshold or post-processing choice was made from test scores; the seed ensemble is uniform, not best-of-three.
- **Locking with hashes.** `selection_lock.json` freezes artefact hashes, development tensor hashes and the baseline checksum; the runner revalidates checkpoint config, seed, 40 completed epochs and protocol before reuse.
- **Reduced smoke runs cannot unlock** a test evaluation.

What it cost

- **36 final neural runs completed** (6 methods × 3 seeds × 2 datasets), plus 15 descriptor candidates and an earlier validation-only sweep kept for provenance.
- Summed per-run training time: 287.19 s (HAR) and 225.19 s (WISDM) on an RTX 5070 Ti, both sweeps running concurrently.
- GPU training is not promised bitwise reproducible; the real TF32 check observed zero embedding difference with cuDNN TF32 disabled, and a preliminary TF32-enabled run exceeded tolerance, so that path is rejected rather than silently accepted.

The negative result, kept in view

HAR hybrid won validation at 3.04% EER and reached 21.46% on test — worse than the 17.64% baseline it replaced.

- Per-seed test EER 0.2130 / 0.2004 / 0.2288, ensemble 0.2146.
- The same pattern on WISDM: validation 1.98% → test 9.58%.
- It is reported in full in the repository, and no later candidate was chosen from test scores to replace it.

03 · ACTIVITY CLASSIFICATION

Six activity classes, and the selection rule that changed the numbers

Activity recognition is the first stage of the pipeline: only walking windows are used for identity comparison. The five-track sweep re-ran every architecture for 20 full epochs and selected each checkpoint on **validation activity loss** instead of training loss — a stricter rule that lowered several reported accuracies. The historical numbers are shown alongside for comparison, and the difference is reported rather than smoothed over.

Test activity accuracy by track

Solid bars: current validation-selected sweep. Muted bars: historical run selected on training loss. Dashed line: best historical single run (0.9369, four validation subjects).

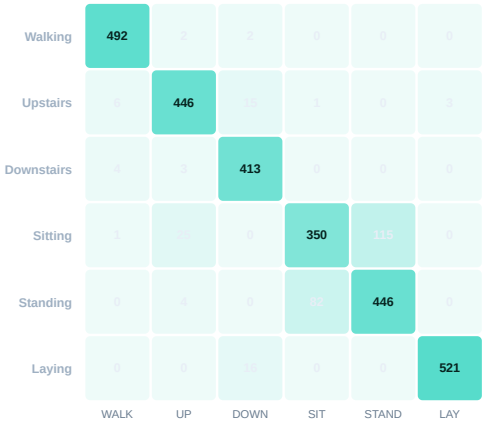


- validation-selected sweep (current protocol)
- historical, train-loss selected
- historical best single run (different protocol)

WISDM activity track: 86.68% test accuracy for the same DeepConvLSTM, on a different dataset with different class balance. It is a separate measurement, not a second view of the HAR number.

Confusion matrix — best current track (multitask, 90.53%)

Counts of 2,947 test windows. The remaining error is concentrated in *standing* ↔ *sitting* — a posture confusion, not a gait one.



Rows = true class, columns = predicted class. Hover any cell for the row share.

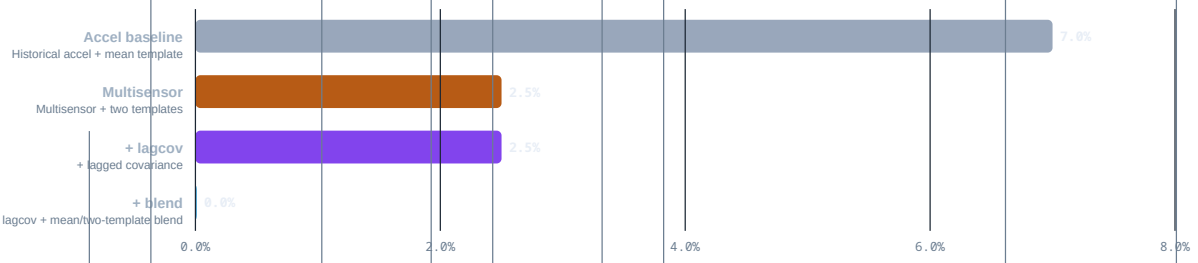
04 · CROSS-DAY & MULTISENSOR

The frozen cross-day error fell to zero — and the hard user was fixed

Santos v7 is controlled leg-mounted walking collected on two different days. The overnight hillclimb changed the representation (accelerometer + gyroscope + rotation-invariant descriptors), the backend (lagged covariance), and the enrollment scorer (a mean/two-template blend). Each change was selected on development folds, and its threshold was calibrated on five separate CALIB users before test access. The result below is the first time the pipeline separates all five test users across days — with an important caveat: the test cohort is the **same five people already inspected** in the earlier study, so this is exploratory, not a fresh blind evaluation.

Cross-day error by method (frozen-threshold test)

Each bar is the same five test users, day-1 enrollment, day-2 probing. EER is independent of the threshold; AUC and the fixed FAR/FRR are shown in the table below.

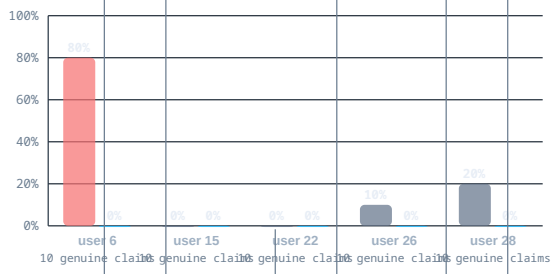


METHOD	EER	AUC	FALSE ACCEPTS	GENUINE REJECTIONS
Historical accel + mean template	7.0%	0.9888	1/200 (0.5%)	11/50 (22%)
Multisensor + two templates	2.5%	0.9932	6/200 (3.0%)	1/50 (2%)
+ lagged covariance	2.5%	0.9958	7/200 (3.5%)	0/50 (0%)
lagcov + mean/two-template blend	0.0%	1.0000	3/200 (1.5%)	0/50 (0%)

The zero rejections are real for these five people, but the false-accept rate still exceeds target. The frozen threshold yields 1.5% test FAR against a 1% goal, and six of the seven false accepts across the lagcov variants target identity 26. Perfect separation of a small reused score set does not establish perfect authentication, and no threshold was re-chosen from the test gap.

The hard user, before and after

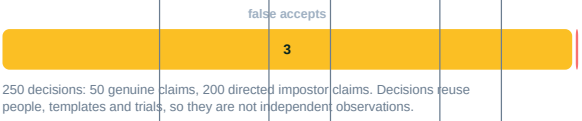
False-reject rate per test user. Muted bars: the old accel baseline. Solid bars: the lagcov + blend method. User 6 — refused 8 of 10 claims before — now passes all ten.



The 22% aggregate FRR was never a population average. It was one person at 80% and everyone else near zero. The lagcov backend pulled user 6 into full separation, which is why the aggregate moved to zero.

Error budget on the selected threshold

Threshold −0.7250201527062856, accept iff `score ≥ threshold`, calibrated on CALIB users only and never changed after test scoring.



250 decisions: 50 genuine claims, 200 directed impostor claims. Decisions reuse people, templates and trials, so they are not independent observations.

TEST RESULT

0.0% EER

ROC AUC 1.0 · FAR 1.5% (3/200) · FRR 0% (0/50)

FROM THE OLD BASELINE

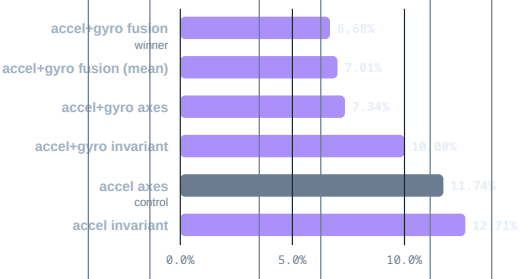
7.0% EER

FAR 0.5% (1/200) · FRR 22% (11/50) · one user refused 8/10

What this still does not establish. Five reused test identities; no independent confirmation; controlled leg-mounted sensor, not a phone; no confidence interval; no takeover-delay measurement. The FAR of 1.5% is above the intended 1%.

Multisensor representation: fusion wins development

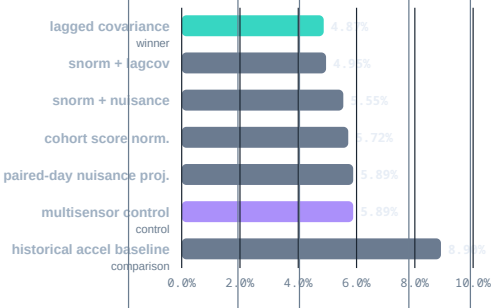
Thirty sensor/representation/enrollment candidates on five development subject folds. Only the key arms are shown; the full thirty remain in the evidence. Mean development EER, lower is better.



Accelerometer + gyroscope axes with rotation-invariant descriptors, training-only WCCN(automatic), two angular medoid templates scored by maximum cosine. The winner needed no final neural training.

Backend: lagged covariance wins

Six backends over twenty unique held-out triples (four balanced five-fold partitions). Mean EER, lower is better. The historical baseline is shown for reference.



Lagged covariance (362-dimensional descriptor) won at 4.874% versus 5.885% for the multisensor control and 8.897% for the historical accel baseline under the same twenty splits.

What this study does establish. Cross-day verification is *measurable* and *improvable* under a frozen protocol: a change to representation and backend moved the same five test users from 7.0% to 0.0% EER and from 11 to 0 genuine rejections, without re-tuning the threshold on test data. Selection happened on development folds and separate calibration users.

Why it is not yet a deployment claim. The test cohort is the same five people seen before, so the improvement is partly a confirmation that the new features separate an already-observed population — not proof of generalisation to unseen people. The false-accept rate is still above target. Fresh repeated-session phone data remain the blocker.

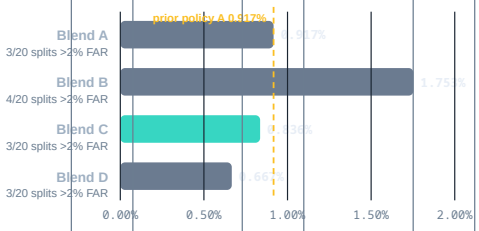
05 · CALIBRATION

A pooled threshold and a twelve-person refit lower the development false-accept rate

Separate from the hillclimb, this DEV-only study asks whether the operating point can be chosen more reliably. Twenty outer validation triples come from four balanced five-fold partitions of fifteen development people; each split holds out three people for calibration and three for validation, and fits the metric on the remaining nine (or twelve, for the refit). The promotion gate was fixed before fitting: blend FAR ≤ 1%, no increase in splits above 2% FAR, strictly lower mean FRR, and at least ten people no worse.

Calibration policy comparison (blend scorer)

Mean FAR and mean FRR across twenty development splits. Policy C — the pooled threshold applied to a twelve-person refit — was the sole eligible replacement.



mean FAR (lower is better)

selected policy C

What policy C changes, and what it does not

MEAN FAR

0.917% → 0.836%

97/11,600 vs 11/1,200 exact

MEAN FRR

19.28% → 15.13%

329/2,175 vs 559/2,900 exact

The concentrated-false-accept diagnostic triggered. Eight of the prior blend's eleven development false accepts sit in two directed person pairs. This flags claimant normalization as the next bounded research question; no normalization arm was added here, and those repeated decisions are not eleven independent people.

Per-person limits remain. Five people improve, seven tie, three worsen under C. Identity 9 is still completely rejected (100% mean FRR) and identity 5 stays at 67.5%. Passing the aggregate gate does not establish usable authentication for everyone.

What it establishes. A pooled threshold plus a twelve-person refit passes a fixed promotion gate

What it does not. These are descriptive results from overlapping development splits, not new

on development data: lower mean FAR and FRR with no increase in bad-split count. Exact rational arithmetic is used throughout, and no original CALIB or TEST numerical payload was opened — this study neither changes the earlier frozen threshold nor establishes a better TEST FAR.

held-out or population estimates. The order-statistic reference policy (D) assumes independent calibration observations, which does not hold for dependent pooled gait scores. Two hundred historical CPU covariance fits were run in total after a discarded first attempt.

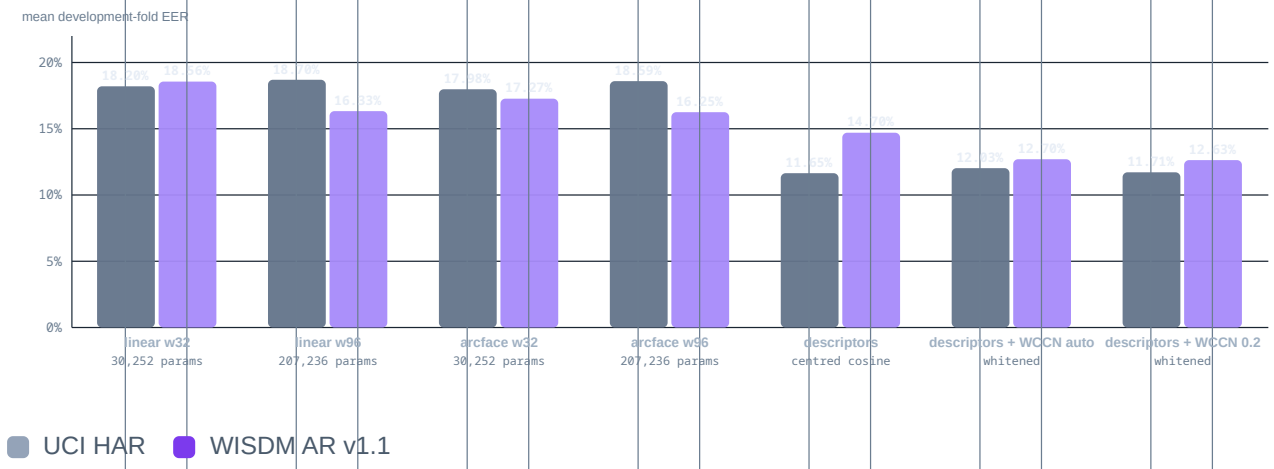
06 · SUBJECT GENERALISATION

Bigger encoders did not help; simpler features did

Forty-eight complete runs tested a matched size × objective factorial on subject folds rather than random windows: width 32 vs 96, linear vs ArcFace identity heads, three subject folds × two seeds × exactly 40 epochs per neural run, with the source test split never loaded. Larger encoders helped WISDM slightly and made HAR slightly worse, and both lost to descriptor baselines.

Mean development-fold EER by candidate

Whiskers are fold standard deviation (dependent fold variation, not a confidence interval). Development folds only — the source test split was never loaded in this study.



Relative development error reduction

Against the small linear-head comparator, on the same folds.

UCI HAR

−36.0%

18.203% → 11.647%
(features_standard)

WISDM V1.1

−31.9%

18.560% → 12.633%
(features_wccn_0.2)

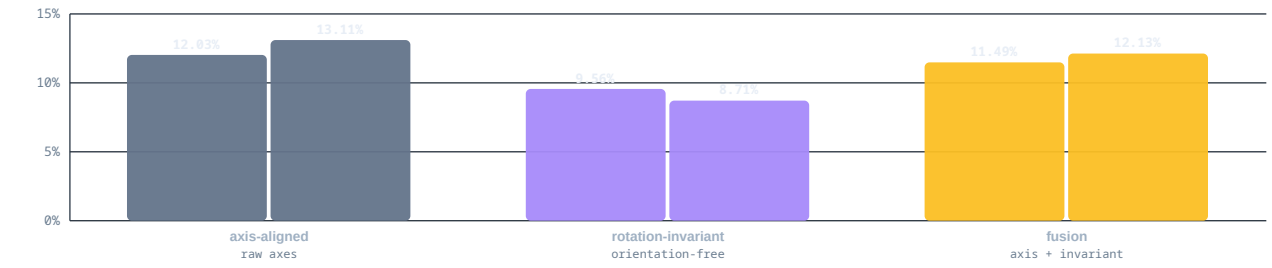
> **Read this study as a development result.** Earlier work already inspected the HAR and WISDM test cohorts; this work does not relabel them as fresh. Four users per source are excluded from every optimizer and reporting fold, and calibration happens after selection.

Why this matters for the next step

- **Descriptor methods beat larger neural encoders on both sources** when the problem is framed as subject generalisation rather than window discrimination.
- The same descriptor family (`wccn_auto`) is what won the locked cross-day study — the two independent lines of evidence point the same way.
- **Careful:** these are development folds. Fold SDs of 6–10 percentage points mean the ranking is fragile and the margins between the top candidates are small.

Rotation-invariant descriptors transfer to HAR and WISDM

A development-only invariance ablation: replacing axis-aligned descriptors with rotation-invariant ones improves mean subject-fold EER on both sources. Within-recording subject folds, not fresh cross-day or TEST results.



axis-aligned rotation-invariant fusion (axis + invariant)

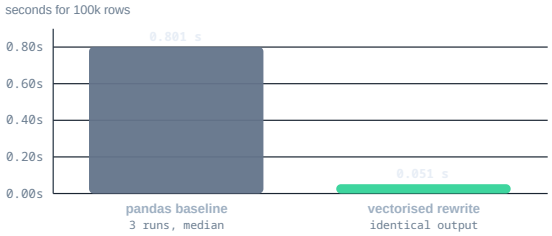
07 · COST

What it costs to run

Data loading was rewritten for throughput, inference latency was measured on CPU and GPU, a compression experiment was run on the activity model, and the multisensor cross-day winner was timed against its baseline. Two of these results are clean wins; the compression experiment is a documented failure, and the accuracy gain has a measured latency cost.

Preprocessing: 15.72× faster, identical output

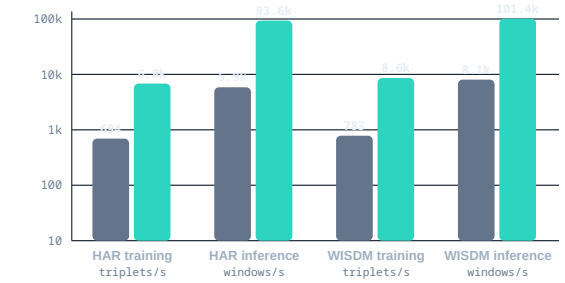
100,000 rows, window 128 / stride 64, three runs each, medians shown. Output equality was checked, not assumed.



The rewrite is a vectorised load path over the same windows; the produced windows are bit-identical to the pandas baseline.

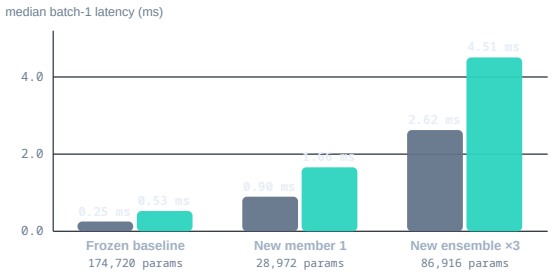
Training and inference throughput

Log scale. CPU on two threads, CUDA on an RTX 5070 Ti; warm-up batches excluded, medians over repeated batches.



Inference latency vs model size

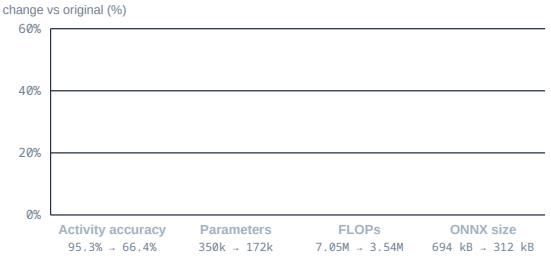
Resident input, FP32, five warm-up batches, 30 timed repetitions. Excludes input transfer, data loading and scoring.



Fewer parameters, more time. The new ensemble stores 86,916 parameters against the baseline's 174,720 (the old count includes an unused DAE decoder) but is slower, because many small multiscale operations and three forwards add dispatch cost. Optional branch packing cuts CUDA batch-1 latency from 4.693 ms to 3.687 ms (1.27×) while growing the stored ensemble to 126,084 parameters and slowing CPU batch-128 from 76.5 ms to 92.8 ms. **No blanket speed-up, phone-energy or on-device claim is made.**

Compression experiment — a documented failure

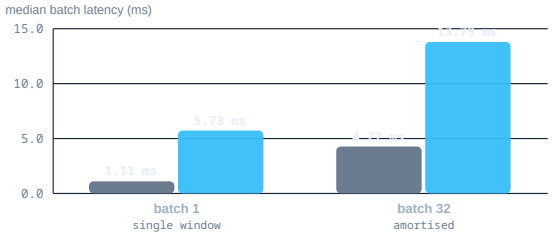
Pruning the activity model bought size and cost accuracy. The accuracy loss was far too large to ship, so the branch is reported, not adopted.



Test activity accuracy fell from 95.31% to 66.41% (−30.3%) while parameters halved and FLOPs dropped by half. The size win is real; the accuracy loss is not acceptable, so the compressed model is not part of the pipeline.

Multisensor winner: an accuracy/latency trade-off

Resident raw-window-to-embedding CPU latency, batch one, thirty timed repetitions on two CPU threads. The cross-day winner is measurably slower than the acceleration baseline it replaces.



5.2× slower at batch one, not a speed-up. The multisensor winner needs 5.73 ms per window against the accel baseline's 1.11 ms (batch-32: 13.79 ms vs 4.27 ms). This is the measured cost of the accuracy gain, not an execution optimisation. Feature extraction, fixed projection and normalization are timed; sensor acquisition, file loading, enrollment/claim scoring and phone energy are excluded.

08 · INTERPRETATION

What this project shows, and what it does not

The honest summary of a project at this stage: the pipeline works, the protocol is disciplined, cross-day verification improved dramatically overnight, and the deployment claims are still not earned — the biggest gains reuse a small already-inspected cohort.

Supported by the evidence

- WISDM walking verification error more than halved (20.10% → 9.58%) under a validation-locked, test-read-once protocol, with ROC AUC 0.9622.
- Cross-day verification on the same five test users moved from 7.0% EER / 22% FRR to 0.0% EER / 0% FRR with a multisensor + lagcov + blend pipeline — the formerly-refused user now passes all ten claims.
- A pooled-threshold calibration policy (C) lowered development mean FAR from 0.917% to 0.836% and mean FRR from 19.28% to 15.13% under a fixed promotion gate.
- Subject-fold studies consistently favour descriptor methods over larger neural encoders, and rotation-invariant descriptors transfer to both HAR and WISDM.
- Data loading is 15.72× faster with bit-identical output; training and inference scale to a single consumer GPU.

Not yet established

- Phone-based or wearable-based authentication of any kind.
- Continuous passive authentication, online takeover detection or a usable false-accept operating point (the best cross-day FAR is still 1.5%, above the 1% target).
- Generalisation beyond the five cross-day test users — the 0.0% EER result reuses the same people inspected earlier, so it is not a blind evaluation.
- A fresh held-out cohort for the exploratory HAR descriptor result (10.44% EER), which reused a test cohort that had already been inspected.
- Any on-device latency or energy claim; the multisensor winner is 5.2× slower than baseline at batch one.

Known weaknesses to fix first

- **The frozen false-accept rate still exceeds target.** Six of seven lagcov false accepts target identity 26, and no threshold was re-chosen from the test gap.
- **Calibration per-person failures persist.** Identity 9 is still 100% rejected and identity 5 stays near 67.5% under policy C — the aggregate gate hides unusable individuals.
- The HAR neural winner generalised worse than the baseline it replaced, and the ensemble's advantage is unproven on that dataset.
- Random-pair EER is not an authentication protocol; enrollment and session behaviour are only approximated.
- Legacy numeric WISDM CSV labels remain unnamed and unverified; only the explicit `wisdm_v1` mapping to the official AR v1.1 release is verified.
- Overlapping windows are not independent, and pair draws use replacement — variance estimates are descriptive only.

Recommended next steps

Ordered by what unblocks the most, not by what is most interesting.

1 **Confirm the cross-day win on a fresh cohort**

The 0.0% EER reuses five already-inspected users. Recruit a new repeated-session phone cohort to turn the exploratory result into a blind one — this is the single highest-value next step.

2 **Study claimant normalization**

Eight of eleven development false accepts concentrate in two directed person pairs, and six of seven lagcov test accepts target one user. A normalization arm is the next bounded research question.

3 **Fix the calibration per-user floor**

Identity 9 is still 100% rejected and identity 5 stays near 67.5% under policy C. Until a per-user floor exists, continuous authentication is not deployable for everyone.

4 **Reproduce the merged pipeline locally**

Pull the merged branches and re-run the full pipeline end to end, so the WISDM identity handling and the cross-day numbers are independently reproduced rather than inherited from pull requests.

5 **Regenerate processed WISDM with subject IDs**

Re-export WISDM with subject identifiers attached, verify the old numeric labels' semantics, and record session boundaries instead of inferring identity from row order.

6 **Decide the target device**

Phone, wrist-worn or ankle-worn changes the windowing, sensor noise and energy budget — and the winner is already 5.2× slower. Optimising latency before that decision is wasted effort.

▸ **Methodology notes and protocol details**

▸ **Source artefacts and how to re-read them**

Prepared for the minor specialisation project team · source `VariableThe/gait-biometrics` @ `4014d85` · generated 24 October 2026

Datasets: UCI HAR, WISDM AR v1.1 (official release, archive SHA256 `e5737912…592d7`), Santos v7 leg-mounted protocol. All numbers are measured values from committed artefacts, not projections.

Negative results, exploratory labels and protocol caveats are reproduced here as they appear in the source reports.